

Kavindu Peiris

Software Engineering Undergraduate

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Summary

Software Engineering undergraduate with hands-on experience building enterprise desktop applications, full-stack web platforms, AI-powered solutions, and cybersecurity systems for academic and real-world clients. Skilled in developing scalable backend systems, RESTful APIs, modern web applications, and machine learning solutions using React, Node.js, Python, Flask, SQL, and PostgreSQL. Published researcher passionate about software engineering, artificial intelligence, cybersecurity, and secure system design.

Education

BComp (Hons.) in Software Engineering , <i>University of Sri Jayewardenepura</i> Expected Graduation: 2027 Current GPA: 3.30/4.00	2023 – Present
Diploma in Information Technology & English , <i>ESOFT Metro Campus</i>	2022 – 2023
G.C.E. Advanced Level (Physical Science Stream) , <i>Prince of Wales' College, Moratuwa</i>	2021 – 2022

Skills

Programming — Java | JavaScript | TypeScript | Python | C# | C/C++ | SQL | Dart
Frontend — React | HTML5 | CSS3 | Tailwind CSS | Streamlit
Backend — Node.js | Express.js | Flask | FastAPI | ASP.NET Core | REST APIs
Database — PostgreSQL | MongoDB | MySQL | SQLite | Prisma ORM
AI & Machine Learning — Scikit-learn | TensorFlow | PyTorch | SHAP | LIME | FAISS
Data Analytics & Visualization — Pandas | NumPy | Chart.js | Matplotlib | Seaborn
Tools & Technologies — Git | Docker | Nginx | Electron | Flutter | JWT Authentication | Postman
Soft Skills — Client Communication | Leadership | Problem Solving | Analytical Thinking | Team Collaboration
Professional Experience — Real Client Project Delivery | End-to-End Project Development

Projects

Enterprise Sales & Field Force Management System 🌐, <i>Custom Desktop Application for Bio Care Consumers, Panadura</i> Developed an enterprise desktop application to streamline field sales, inventory distribution, customer collections, and business reporting. Implemented offline-first architecture with KPI dashboards, authentication, route management, and automated database backup. <i>Tech Stack: Electron • React.js • Node.js • SQLite3 • JavaScript • Tailwind CSS • Recharts</i>	01/2026 – 05/2026
Enterprise Stock Control & Inventory Management System 🌐, <i>Custom Enterprise Desktop Application for JFI CONSOLIDATED (PVT) LTD</i> Developed a custom enterprise desktop inventory management system to digitize warehouse operations, purchasing, supplier management, and stock tracking. Implemented analytics dashboards, reporting, authentication, and automated database backup in an offline-first architecture. <i>Tech Stack: Electron • React.js • Node.js • SQLite3 • JavaScript • Tailwind CSS • Recharts</i>	08/2025 – 12/2025

Student Success Platform  , <i>Cloud-Native Microservices SaaS Academic Management System</i> Developed a cloud-native microservices-based SaaS platform for academic management, GPA tracking, and performance analytics with secure REST APIs and Dockerized deployment. <i>Tech Stack: React.js • TypeScript • Node.js • Express.js • PostgreSQL • Prisma ORM • JWT • Docker • Nginx • REST APIs • Microservices</i>	05/2026 – 07/2026
AI Workforce Intelligence Platform  , <i>Machine Learning & Explainable AI Workforce Analytics System</i> Developed an AI-powered workforce analytics platform that predicts employee layoff risk using Machine Learning, Deep Learning, and Explainable AI. <i>Tech Stack: Python • Flask • Streamlit • CatBoost • TensorFlow • Scikit-learn • LIME • SQLite</i>	03/2026 – 05/2026
LectureMind AI  , <i>Local RAG-Powered AI Study Assistant</i> Developed a privacy-first RAG-powered AI study assistant that generates notes, flashcards, quizzes, and answers questions from lecture materials using local LLMs. <i>Tech Stack: Python • Streamlit • FAISS • FLAN-T5 • Sentence Transformers • SQLite • PyTorch</i>	04/2026 – 06/2026
ThreatOps  , <i>AI-Powered Insider Threat Detection Platform</i> Developed an AI-powered insider threat detection platform for real-time user behavior analytics, anomaly detection, and security monitoring. <i>Tech Stack: React.js • FastAPI • Python • PostgreSQL • Scikit-learn • Docker • WebSockets</i>	05/2026 – 06/2026
CoordBot AI Guard for Coordinated Bot Detection (Final Year Research Project)  , <i>Lightweight Web Behavioural Fingerprinting Framework for Coordinated Bot Detection</i> Developing a real-time behavioural bot detection framework using behavioural biometrics, coordination analysis, and lightweight machine learning for web applications. <i>Tech Stack: Python • Google Apps Script • XGBoost • Machine Learning • Cybersecurity</i>	03/2026 – Present

Publications

A Systematic Review of Coordinated Bot Detection Techniques in Real-Time Web and Mobile Applications <i>J'PURA Computing Symposium (JCoS) 2026 Published & Presented</i>	20/05/2026
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Certificates

- Fundamentals of Machine Learning & Artificial Intelligence - AWS - AI/ML Engineer Stage 2 - SLIIT - AWS Cloud Practitioner Essentials - AWS	- RAG, AI Apps, and AI Agents for Cybersecurity and Networking - LinkedIn Learning - Introduction to Cybersecurity - Cisco Networking Academy - Python Programming - University of Moratuwa
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Interests

- Software Development - Full-Stack Development	- Artificial Intelligence & Machine Learning - Cloud Computing	Cybersecurity	Data Analysis
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References

Mr. Pubudu Jayasekara, *Lecture*, University of Sri Jayewardenepura
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Mr. Asanka De Silva, *Operations Manager*, eNerds
easankad@gmail.com, +94 718622650